

NICHOLAS BRIAN ANYA

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Summary

IT professional with experience across network engineering, telecommunications, and applied data science, including machine learning, NLP, and cybersecurity. Skilled in building and operating scalable technical infrastructure, and in translating complex data into actionable insights for stakeholders.

Education

Master of Science in Information Technology **Jul 2024 – May 2026**

Carnegie Mellon University, Pittsburgh, PA, USA

Concentrations: Data Science, AI/ML, Cybersecurity, Internet Measurement, and Privacy Policy in Tech

Teaching Assistant: Python for Data Analytics

Bachelor of Science in Telecommunications Engineering **Aug 2017 – May 2022**

Makerere University, Kampala, Uganda

Concentrations: Telecommunications Systems (Satellite, Optical, Mobile, Networks, and Antenna Design)

Professional Experience

Graduate Research Assistant, ECE Department **Jun 2025 – May 2026**

Carnegie Mellon University

Research Assistant in Professor Theophilus Benson's Internet Measurement Lab, combining Network Engineering skills and Data Science to uncover insight to improve global internet infrastructure.

- Analyzed multiple raw data sources through custom inference algorithms to infer critical internet infrastructure resilience, particularly sub-sea cables, Internet Exchange Points, and latency overhead resulting from suboptimal routing.
- Contributed to deployment of 40 measurement probes across Rwanda and Senegal, supporting data collection to support resilience assessments of critical government and utility infrastructure.
- Collaborated with industry researchers from Cloudflare, IPinfo, and Hurricane Electric, located in different continents to enrich datasets and improve accuracy of inference algorithms.
- Authored work presented and published at peer-reviewed publications at HotNets'25 and ANRW'25, top venues for applied networking research.

IT Support Technician **September 2024 – Jun 2025**

Carnegie Mellon University

Worked with the university IT department to ensure availability of IT services on campus.

- Managed tickets raised on IT issues across hardware and software support needs.
- Managed audio and visual systems for physical and remote sessions as well as ShareLink setup for wireless presentations.
- Setting up and supporting on-campus local area network including wireless network management and troubleshooting

Network Engineer **April 2022 – Jul 2024**

Research and Education Network for Uganda (RENU), Kampala, Uganda

Network Engineering at ISP level with customer base of Uganda's national research and education community, serving 400+ member institutions and 100,000+ end users across the country's academic and research community.

- Designed and operated routing and switching infrastructure across 400+ integrated sites nationwide, supporting multivendor environments across universities, schools, and hospitals.
- Managed upstream peering as well as downstream customer support including LAN support as well as wireless and fibre optic last-mile solutions.
- Conducted network monitoring and traffic analytics to optimise bandwidth allocation, informing capacity planning decisions that sustained connectivity for 100,000+ end users.
- Delivered a 106-school rural connectivity programme in partnership with the Uganda Communications Commission,

enabling research-grade Internet access for students and educators in underserved communities.

- Led deployment of solar-powered wireless routers and engineered enhanced LTE solutions for rural connectivity, extending eduroam SSID coverage beyond campuses across Uganda.
- Trained staff across member institutions on first-level network troubleshooting, building distributed technical capacity and reducing dependency on centralised support.

Conferences and Publications

- “A Call to Arms: Motivating an Internet Measurements Observatory for Africa.” HotNets '25, Washington D.C., USA, 2025. <https://doi.org/10.1145/3772356.3772405>
- “Towards Comprehensive Mapping of Africa’s Internet Infrastructure.” ANRW, Madrid, Spain, 2025. <https://doi.org/10.1145/3744200.3744768>
- “Bridging the Connectivity Gap Using Unconventional Last-Mile Solutions.” Ubuntunet Connect, Kampala, Uganda, 2023. <https://doi.org/10.60763/africarxiv/52>

Skills

Network Engineering

Routing & switching (BGP, MPLS, IS-IS, OSPF), wireless network design (including 802.1X), fiber optic network design & planning, LAN/VLAN architecture, Firewall configuration, last-mile connectivity (LTE, microwave, fibre), multivendor environments

AI & Machine Learning

PyTorch, TensorFlow, scikit-learn, supervised & unsupervised learning, neural networks (RNN, CNN), model training, evaluation, deployment & interpretability, bias-variance tradeoff, NLP, Hugging Face, BERT/Transformers, fine-tuning, prompt engineering, text classification, sentiment analysis, computer vision

Data Science & Analytics

Python (pandas, numpy, Matplotlib, Seaborn, BeautifulSoup), SQL, Microsoft Excel, Power BI, Jupyter Notebook, EDA, statistical inference, hypothesis testing, data visualization, web scraping, linear algebra, vectorization

Cybersecurity

Ethical penetration testing, cloud security, information security, web application vulnerability assessment (OWASP Top 10)

Privacy Policy in Tech

Privacy harm assessment, Differential Privacy, GDPR and CCPA

Internet Measurement & Research

BGP MRT archives, route collectors, CAIDA, M-LAB, RIPE Atlas, traceroute-derived datasets, Internet infrastructure mapping, policy-relevant peer-reviewed evidence generation

Tools

Git/GitHub, LaTeX, draw.io/Visio, Microsoft Office, Azure, AWS, Linux, English (fluent)

Certifications

- HCIP-Routing & Switching, Huawei.
- JNCIA-Junos, Juniper Networks.
- Advanced BGP Training 2024 - APNIC, Bangkok, Thailand